Phase 3

Final Documentation

Course Name	Network Security
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1. Introduction & Project Scope

Our architectural execution delivered a fully operational, multi-layer defense system:

OPNsense acting as a robust edge security layer to secure the perimeter and establish clear network layering, a specialized DMZ zone successfully integrating advanced monitoring and detection tools like the Cowrie Honeypot and Splunk SIEM, and an internal network segmented across our defined organizational VLANs.

In this phase, our ultimate objective is to ensure that every security principle we intended to apply in the previous phases is accurately reflected, fully functional, and verified within the live environment.

2. Overall Evaluation

The evaluation was conducted by executing a controlled attack scenario against the live GNS3 topology, that will be explained in detailes on the next few pages, each defensive layer was observed, and its behavior was recorded in Splunk for analysis.

2.1 OPNsense Firewall Evaluation

Rule	Intended Action	Test Result
WAN -> Internal (all)	Block all unsolicited inbound	Attacker blocked at perimeter
DMZ -> Internal VLANs	Block all pivot attempts	All pivot pings timed out
Internal VLANs -> DMZ	Allow HTTP/HTTPS/Mail ports only	Only allowed ports responded
Internal -> Internet (NAT)	Masquerade outbound traffic	External connectivity confirmed
Kali Manager -> OPNsense	HTTPS (443) only	Web UI accessible; other ports rejected
IT VLAN -> Management Zone	Allow ports 8000 and 1812	Splunk and AAA reachable from IT only

2.2 Snort IPS Evaluation

Snort was deployed in inline bridge mode (br0), physically positioned between OPNsense and the DMZ switch. This ensured that all DMZ-bound traffic was inspected before reaching DMZ zone.

We know that this is the last phase, but honestly, our last step here is we are working on sending the already to Splunk via UDP syslog (port 514), where they should be indexed and searchable within seconds of generation.

2.3 Cowrie Honeypot Evaluation

Cowrie was configured to listen on port 2222, with an iptables redirect routing all port 22 traffic to 2222 transparently. The attacker will be unaware of this redirection.

The honeypot successfully deceived the attacker throughout the entire session. The attacker remained uncertain whether they were in a real environment, which kept them engaged longer and produced a richer intelligence log.

2.4 Splunk SIEM Evaluation

Splunk was configured to receive UDP syslog on port 514. Log sources included OPNsense firewall logs, Snort alert output (working on this last step as stated on the snort evaluation section), and Cowrie session logs.

3. Risk Assessment

3.1 Risk Register

Risk Description	Affected Zone	Level
SSH brute force on DMZ	DMZ	HIGH
Attacker pivoting from DMZ to Internal	Internal VLANs	HIGH
Unauthorized access to Management Zone	MGMT Zone	MED
VLAN hopping attack	Access Layer	MED
Snort rule evasion (fragmented packets)	DMZ	MED
Splunk log tampering if locally accessible	MGMT Zone	LOW
Honeypot fingerprinting by skilled attacker	DMZ	MED
VPN credential brute force	WAN / VPN	HIGH

3.2 Improvement Strategies

Based on the risk assessment findings, the following improvement strategies are proposed:

Risk	Current Mitigation
SSH Brute Force	Cowrie honeypot traps attacker; Snort alerts
DMZ Pivot Attempt	Firewall blocks all DMZ to Internal traffic
MGMT Zone Access	IT VLAN only, ports 8000 and 1812
VLAN Hopping	Built-in GNS3 switch with explicit port assignments

Snort Rule Evasion	Inline IPS mode drops matched signatures
VPN Credential BF	OpenVPN with certificate + password auth

4. Configuration Details Final Phase Additions

That is what has been added to configuration which was stated on Phase 2, because back then, the attack was not so clear on our vision

4.1 Snort IPS Custom Detection Rules

The following rules were added to the Snort configuration to cover the attack scenario:

```
# Rule 1: SSH Brute Force Detection
alert tcp $EXTERNAL_NET any -> $HOME_NET 22 (\
  msg:"ET SCAN Potential SSH Brute Force Attempt";\
  flags:S; threshold: type threshold, track by_src, count 5,
seconds 60;\
  sid:1000001; rev:1;)

# Rule 2: Nmap SYN Stealth Scan
alert tcp $EXTERNAL_NET any -> $HOME_NET any (\
  msg:"ET SCAN Nmap SYN Scan Detected";\
  flags:S; threshold: type threshold, track by_src, count 20,
seconds 5;\
  sid:1000002; rev:1;)

# Rule 3: ICMP Sweep / Ping Flood
alert icmp $EXTERNAL_NET any -> $HOME_NET any (\
  msg:"ET SCAN ICMP Sweep Detected";\
  itype:8; threshold: type threshold, track by_src, count 10,
seconds 5;\
  sid:1000003; rev:1;)

# Rule 4: Reverse Shell Attempt (TCP callback)
alert tcp $HOME_NET any -> $EXTERNAL_NET 4444 (\
  msg:"ET MALWARE Reverse Shell Callback Detected";\
  flow:established,to_server; sid:1000004; rev:1;)
```

These rules were loaded on the br0 bridge interface, ensuring every packet traversing the DMZ path is inspected.

5.2 OPNsense Log Forwarding to Splunk

Remote syslog forwarding was configured in OPNsense under

System -> Settings -> Logging -> Remote, with the following parameters:

- Transport: UDP(4)
- Hostname: 192.168.100.10
- Port: 514
- Description: Splunk Core SIEM Syslog

This configuration was validated by confirming OPNsense firewall block events appeared in Splunk's index=main within seconds of triggering a blocked connection attempt.

5.3 Splunk SIEM -- Data Input Configuration

Splunk was configured to receive logs via UDP syslog on port 514 under Settings -> Data Inputs -> UDP:

- UDP Port: 514
- Source type: syslog
- Index: main
- Status: Enabled

Splunk enterprise version 9.2.1 was used. The web interface was confirmed accessible at http://192.168.100.10:8000 from the Puppy Linux management node (192.168.100.20).

5.4 VPN Gateway Configuration

An OpenVPN server instance was configured on OPNsense to allow secure remote access for legitimate users to reach internal VLANs only. Key configuration parameters:

Parameter	Value
Role	Server
Protocol	UDP4
Port	1194
Bind Address	10.10.10.1 (WAN interface)

Tunnel Network	10.8.0.0/24
Local Network (Accessible)	192.168.20.0/24, 192.168.21.0/24, 192.168.22.0/24, 192.168.23.0/24
Certificate Authority	VPN_CA (internally generated, SHA256, RSA 2048)
Server Certificate	VPN_Server_Cert (signed by VPN_CA)
Authentication	Local Database (username: vpnuser)

6. Attack Simulation

A controlled, full-lifecycle attack simulation was executed against the live topology to validate the defensive infrastructure and demonstrate the core honeypot principle. The attacker was an Ubuntu Server VM (10.10.10.2) connected to the WAN side of OPNsense via the GNS3 Multiplexer.

6.1 Attacker Machine Setup

The attacker machine was configured as follows:

- Base OS: Ubuntu Server 22.04 (Full Clone of the Golden Image)
- IP Address: 10.10.10.2/24, Gateway: 10.10.10.1 (OPNsense WAN)
- Tools installed: nmap, hydra, netcat, curl, tcpdump

```
sudo apt update && sudo apt install -y nmap hydra netcat curl tcpdump
```

6.2 Phase Summary

Phase	Name	Key Commands	Detected By
1	Reconnaissance	nmap -sn, nmap -sS -T2, nmap -sV -sC	Snort + OPNsense
2	Brute Force	nc -v 192.168.30.30 22 && hydra -l root -P rockyou.txt ssh://...	Snort SSH rule
3	Post-Exploitation	uname -a, cat /etc/passwd, cat /etc/shadow, arp -a	Cowrie session log
4	Honeypot Detection	free -m, ps auxf, ss -tuln, last, lastlog	Cowrie (all returned fake data)

5	Lateral Movement	ping 192.168.20.1, ssh root@192.168.30.10, bash reverse shell	OPNsense BLOCKED
6	Cover Tracks	history -c, rm -rf /var/log/*, pkill rsyslog	Splunk (evidence preserved)

Outcome: The attacker successfully authenticated into the Cowrie honeypot, executed extensive post-exploitation activity, attempted lateral movement into all internal VLANs, attempted a reverse shell callback, and finally attempted to destroy all evidence. Every action was blocked at the appropriate layer or logged in Splunk. No internal asset was affected.

6.3 MITRE ATT&CK Framework Mapping

Tactic	Technique ID	Tool Used	Detected By
Reconnaissance	T1595 <--> Active Scanning	Nmap	Snort + OPNsense
Initial Access	T1110 <--> Brute Force	Hydra	Snort SSH BF Rule
Execution	T1059 <--> Command & Scripting	bash	Cowrie session log
Persistence	T1098 <--> Account Manipulation	useradd, crontab	Cowrie session log
Credential Access	T1003 <--> OS Credential Dumping	cat /etc/shadow	Cowrie (fake data returned)
Lateral Movement	T1021 -- Remote Services	SSH, ping	OPNsense BLOCKED
Command & Control	T1571 -- Non-Standard Port	nc / bash reverse shell	OPNsense BLOCKED
Defense Evasion	T1070 -- Indicator Removal	rm, history -c	Splunk (already preserved)

7. Testing & Final Notes

7.1 Connectivity Testing

- Ping tests from each VPCS device to their respective L3 switch gateway.
- Ping tests from L3 switches to OPNsense LAN interfaces (10.0.0.1 and 10.0.0.5).
- Ping from Kali Manager (192.168.50.2) to OPNsense (192.168.50.1).

- Ping from Puppy Manager (192.168.100.20) to Splunk (192.168.100.10).
- Ping from OPNsense to DMZ servers (192.168.30.10, .20, .30, .40).
- Cross-VLAN ping from IT VLAN to HR VLAN via L3 switches.
- Ping from DMZ to Internal VLANs: BLOCKED by OPNsense.

7.2 Service Verification

- Apache Web Server: sudo systemctl status apache2 --> Active (running)
- Postfix Mail Server: sudo ss -tuln | grep 25 --> Port 25 LISTENING
- Cowrie Honeypot: bin/cowrie status --> Running (PID confirmed)
- FreeRADIUS AAA: radtest Admin admin123 192.168.100.20 0 radius123 --> Access-Accept
- Splunk SIEM: sudo /opt/splunk/bin/splunk status --> splunk is running
- OPNsense VPN: OpenVPN instance Corporate_VPN --> Active on UDP 1194

8. Security Policy Documentation

8.1 Firewall Change Management Policy

- No rule may be added that permits DMZ-to-Internal or WAN-to-Internal traffic without explicit written approval from the network security administrator.
- Firewall configurations must be backed up after every change using OPNsense's built-in backup utility.

8.2 Logging & Monitoring Policy

- All alerts must forward logs to Splunk (192.168.100.10) via UDP syslog on port 514.

9. Conclusion

We built a complete security network from scratch in GNS3.

We took it from a simple design on paper to a fully working system.

Our testing showed that all our defense tools work perfectly together as one unit. We also looked at potential risks and wrote a clear plan to handle them.

This project covered all learning goals of our course. We put a lot of heart into applying everything we learned this semester. We truly hope our work is good enough and meets your expectations. Thank you